Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Binomial Expansion

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. First three terms of $(1+3x)^4$. **[3]**

2. First three terms of $(1-2x)^5$. **[3]**

3. First three terms of $(1+x)^6$. **[3]**

4. First three terms of $(2+x)^4$. **[4]**

5. First three terms of $(1-3x)^4$. **[3]**

6. First three non-zero terms of $\frac{1}{\sqrt{1+3x}}$. State range of validity. **[6]**

7. First three non-zero terms of $\frac{1}{\sqrt{1-2x}}$. State range of validity. **[6]**

8. First three non-zero terms of $(1+4x)^{-1/2}$. State range of validity. **[6]**

9. First three non-zero terms of $(1-x)^{-1/3}$. State range of validity. **[6]**

10. Coefficient of x^2 in $(1-2x)^5(2+x)^2$. **[5]**

11. Coefficient of x^2 in $(1+x)^4(1-3x)^3$. **[5]**

12. Coefficient of x^3 in $(1+2x)^5(1-x)^2$. **[5]**

13. First three terms of $(1-2x)^{-3}$. State range of validity. **[5]**

14. First three terms of $(1+x)^{-2}$. State range of validity. **[5]**

15. First three terms of $(1-3x)^{-4}$. State range of validity. **[5]**

Section B – Hard (Q16–25)

16. Coefficient of x^2 in $\frac{(1+x)^3}{\sqrt{1-2x}}$. **[7]**

17. Coefficient of x^2 in $\frac{(1-x)^2}{\sqrt{1+3x}}$. **[7]**

18. Coefficient of x^3 in $\frac{(1+2x)^2}{(1-x)^{1/2}}$. **[7]**

19. $(1+ax)^n$ begins $1+12x+66x^2+\cdots$. Find a, n. **[5]**

20. $(1+bx)^m$ begins $1-8x+28x^2+\cdots$. Find b, m. **[5]**

21. Expand $\sqrt{1+2x}$ up to x^3. State range of validity. **[6]**

22. Expand $\frac{1}{(1-x)^2}$ up to x^3. State range of validity. **[5]**

23. Find the term independent of x in $\left(x+\frac{2}{x}\right)^6$. **[5]**

24. Find the term independent of x in $\left(2x-\frac{1}{x}\right)^5$. **[5]**

25. Use binomial expansion to approximate $\sqrt{1.02}$ to 4 d.p. **[5]**

End of Worksheet • *Show all working clearly.*